The observed temperature power spectrum through fourth order in a radiation-only universe: final report

Unified analysis (branch `unified/temperature-spectrum`)

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Abstract

We compute the observed temperature anisotropy through third order, and the angular power spectrum through fourth order, in the primordial amplitude ϵ. The setting is a flat radiation-only universe in general relativity, with Planck test photons emitted on $T_{\rm phys} = T_\star$ and an observer on a fluid worldline.

The calculation uses the fluid's velocity potential and comoving labels as coordinates (*temperature frame*). In these coordinates the observed temperature is an exact line integral of the shear of a conformally rescaled metric, and the observer's own temperature cancels exactly in the normalised map. Further results:

- The spatial Einstein equations take a new mixed trace-free form, which yields a triangular solution scheme with elementary Green functions at every order.
- This scheme satisfies all ten field equations at every order (proved).
- All second-order field kernels are given in closed form for general momenta, in the entire functions Si, Cin and $\mathfrak{e}_m$. This makes every resonance surface manifestly regular.
- Third-order kernels are explicit nested integrals.
- The complete map $\Theta^{(1,2,3)}$ is constructed. It retains photon propagation, emission surface, observer clock, tetrad and normalisation terms.
- $C_\ell^{(11)}$, $C_\ell^{(22)}$ and $C_\ell^{(13)}$ are reduced to exact finite-dimensional integrals, with every sky integral done in closed form.
- In the superhorizon limit the complete observable is obtained in closed form, *exact in* ζ. It gives

$$C_2 = \tfrac{4\pi}{225}\epsilon^2\sigma^2\Xi^2\big[1 + \tfrac{29}{3}\epsilon^2\sigma^2\big] + O(\epsilon^6, k_0^6), \qquad \Xi \equiv x_o^2 - x_\star^2.$$

Every displayed identity is checked by exact computer algebra (no numerics). We state explicitly which integrals remain unevaluated, and which limits remain open.

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1 Problem and summary of results

The task (`problem_statement.tex`) asks for $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ for the observed map

$$\Theta(\hat{n}) = \frac{T_{\rm obs}(\hat{n})}{\frac{1}{4\pi}\int \mathrm{d}\Omega'\, T_{\rm obs}(\hat{n}')} - 1, \qquad T_{\rm obs} = T_\star \frac{E_o}{E_e}, \quad E = -u_\mu p^\mu.$$

The physical model is a perfect radiation fluid with $P = \rho/3$, $\rho = \alpha T_{\rm phys}^4$ and $\Lambda = 0$, with Gaussian adiabatic primordial perturbations and no independent higher-order modes. The calculation must be analytical and keep every term, with all normalisations fixed. A monochromatic specialisation is to be made only at the end.

Requirement	Result (section; status)
Gaussian variable; no higher-order modes	$\zeta = \frac{1}{6}\ln\det\gamma_{ij}$, exactly conserved; Def. 4 (§4)
Constrained initial data at every order	regular solution at any t_i; constraints proved (Thm. 5); forced data nonzero
Field solution through ϵ^3 incl. vectors/tensors	triangular scheme (§5); 2nd order [closed form]; 3rd order [explicit nested integrals]
Observer, clock, tetrad, emission surface	§6; the tetrad reduces to a symmetric root (Prop. 6)
Complete $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$	§6, with an independent generator and Ward checks
$C_\ell^{(11)}$	[closed form] ($\ell = 2, 3$; general ℓ as a 1D integral)
$C_\ell^{(22)}$, $C_\ell^{(13)}$	[exact finite-dimensional integrals], sky integrals evaluated (§8); existence proved

Requirement	Result (section; status)
Monochromatic, held-fixed quantity	$\langle g^2\rangle = \sigma^2$ fixed (§8.2)
Asymptotic limits	superhorizon: [closed form, exact in ζ]; late time: [first order]; general k_0 at fourth order: [open] (§9)
Convergence for general spectra	compact support: [proved]; IR: leading cancellation [derived] from the Ward identities; UV: [power counting only]
Verification	§10 (exact symbolic, generic-jet proofs)

2 Conventions and the experiment

Signature $(-,+,+,+)$, $c=1$. The background is $a = a_1 t$ and $\bar{T} = \mathcal{T}/(a_1 t)$, with $8\pi G\alpha\mathcal{T}^4 = 3a_1^2$.

Fourier and perturbative conventions.

- $g(X) = \int_k g_k e^{ik\cdot X}$, with $\int_k = \int \mathrm{d}^3k/(2\pi)^3$ and $\langle g_k g_{k'}\rangle = (2\pi)^3\delta^3(k+k')P(k)$; also $\mathcal{P}(k) = k^3P/2\pi^2$.
- Every field is $Y = \sum_r \epsilon^r Y^{(r)}/r!$ with $Y^{(r)} = \int_{k_1\ldots k_r} K_r(k_1,\ldots,k_r)\, g_{k_1}\cdots g_{k_r} e^{iK\cdot X}$ and $K = \sum k_a$. The symmetric kernels carry no extra $1/r!$.
- *Multilinear rule:* K_r is the coefficient of $z_1\cdots z_r$ when $g = \sum_a z_a e^{ik_a\cdot X}$ is inserted and only monomials in distinct labels are kept. This fixes every combinatorial factor.
- Spherical harmonics $Y_{\ell m}$ are complex, with the Condon–Shortley phase.

Measurement labels.

- *Emission:* the surface $T_{\rm phys} = T_\star$. Emission is isotropic and Planckian in the fluid rest frame. The intersection is the branch continuously connected to the background one.
- *Observer:* the fluid line $X = 0$. The initial event is $T_{\rm phys} = T_I > T_\star$ on that line, and observation follows after fixed proper time τ_o.
- *Tetrad:* Fermi–Walker transported, with initial orientation $\gamma(t_I,0)^{-1/2}$ (the symmetric square root, aligned with the marked labels).
- *Background times:* $\bar{t}_\star = \mathcal{T}/(a_1T_\star)$, $\bar{t}_I = \mathcal{T}/(a_1T_I)$ and $\bar{t}_o^2 = \bar{t}_I^2 + 2\tau_o/a_1$. We write $x \equiv kt/\sqrt{3} = k/(\sqrt{3}\,aH)$.

3 Exact framework

Lemma 1 (Temperature frame)**.** *For an irrotational radiation fluid, $Tu_\mu = -\mathcal{T}\partial_\mu t$ defines the velocity potential t. In coordinates (t, X^i), with comoving labels X^i,*

$$g = N^2\big(-\mathrm{d}t^2 + \gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j\big), \qquad N = \mathcal{T}/T_{\rm phys}, \qquad u = N^{-1}\partial_t$$

exactly. Euler's equation holds identically, and continuity is equivalent to $\partial_t \det\gamma = 0$. Hence $\gamma = e^{2\zeta(X)}\exp H$ with $\operatorname{tr} H = 0$, and the shear $\sigma_{ij} = \frac{1}{2}\partial_t\gamma_{ij}$ is traceless.

Lemma 2 (Exact redshift)**.** *Null geodesics of g and of $\tilde{g} = g/N^2$ coincide, and $E = \tilde{E}/N$. Hence*

$$T_{\rm obs}(\hat{n}) = T_{\rm fluid}(o)\, e^{-I(\hat{n})}, \qquad I = \int_{t_e(\hat{n})}^{t_o} \sigma_{ij}(t, X(t))\, e^i e^j \, \mathrm{d}t,$$

where $\mathrm{d}X^i/\mathrm{d}t = e^i$, $\gamma_{ij} e^i e^j = 1$ and $\mathrm{d}e^i/\mathrm{d}t = -2S^i{}_j e^j - \Gamma^i{}_{jk} e^j e^k + e^i \sigma_{jk} e^j e^k$, with $S = \gamma^{-1}\sigma$. The emission event is the root of $\ln t_e + \nu(t_e, X(t_e)) = \ln \bar{t}_\star$, with $N = a_1 t e^\nu$.

$T_{\rm fluid}(o)$ is direction-independent. It therefore cancels from Θ *exactly*. The emission-surface temperature shift, Sachs–Wolfe, Doppler and integrated Sachs–Wolfe terms are all contained in the single integral I.

Proposition 3 (Unified field equations)**.** *With $n = \ln N$, Einstein's equations are equivalent to*

$$(N) \quad 3\nu'' - \Delta_\gamma \nu - 2|D\nu|^2 + \operatorname{tr} S^2 + \tfrac{3}{t^2}(e^{-2\nu} - 1) = 0, \tag{1}$$

$$(T) \quad \mathrm{TF}\left[S' + 2(t^{-1} + \nu')S + R^k{}_j[\gamma] - 2D^k D_j \nu + 2D^k \nu D_j \nu\right] = 0, \tag{2}$$

$$(M) \quad D_j S^j{}_i - 2\partial_i \nu' + \tfrac{2}{t}\partial_i \nu + 2S^j{}_i \partial_j \nu + 2\nu' \partial_i \nu = 0, \tag{3}$$

$$(\mathrm{Ham}) \quad R - 4\Delta\nu - 2|D\nu|^2 - \operatorname{tr} S^2 + 6(t^{-1} + \nu')^2 - 6e^{-2\nu}/t^2 = 0. \tag{4}$$

In the mixed form (T), the $\sigma\sigma$ terms of the spatial equation cancel, because $\gamma^{ki}\sigma'_{ij} = (S^k{}_j)' + 2S^k{}_i S^i{}_j$. This is proved for generic γ (six free functions, t-independent determinant) and arbitrary N, on independent symbolic 2-jets.

4 Primordial state and initial data

Definition 4 (Gaussian variable; absence of independent higher-order modes)**.** The state is the solution of Prop. 3 with the following properties:

(G) $\zeta = \epsilon g$ exactly, with g a zero-mean, homogeneous, isotropic Gaussian field;

(T) $\gamma \to e^{2\zeta}\delta_{ij}$ as $t \to 0$. The limit is conformally flat in the marked conformally Cartesian material chart, so there is no tensor seed;

(V) $\mathrm{d}(Tu) = 0$, so there is no vorticity seed;

(R) ν and H are even and analytic in t at fixed momenta, with $\nu(0) = 0$ and $H(0) = 0$, so there are no decaying modes.

"No independent higher-order modes" means $\zeta^{(r\geq 2)} \equiv 0$ together with (T), (V) and (R) at every order.

Because $\det\gamma$ is t-independent, $\zeta = \frac{1}{6}\ln\det\gamma$ at *every* time. As $t \to 0$, constant-t slices become uniform-density slices, so ζ is the nonlinear uniform-density (δN) curvature perturbation.

The residual conformal relabellings are fixed as follows: the dilation by $\langle g\rangle = 0$; special conformal transformations by statistical homogeneity; translations and rotations are statistical symmetries.

Exactly in ζ, the gradient-expansion data are

$$\nu = \tfrac{t^2}{18} e^{-2\zeta}\left(2\nabla^2\zeta + |\nabla\zeta|^2\right) + O(t^4), \quad H_{ij} = \tfrac{t^2}{3} e^{-2\zeta}\ \mathrm{TF}\left[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\right] + O(t^4). \tag{5}$$

The initial data at any t_i are the regular solution of §5 evaluated at t_i, with $\rho = 3a_1^2/(8\pi G N^4)$ and $u = N^{-1}\partial_t$. Higher-order data are nonzero. For example, $\gamma^{(2)}(0) = 4g^2\delta$, and the $\nu^{(2)}$ kernel is $\frac{t^2}{9}[2(k_1^2 + k_2^2) - k_1 \cdot k_2] + O(t^4)$. The forced $K = 0$ mode ($5k^2t^2/9$ at $k_2 = -k_1$) is kept. The data evolve back into the same regular solution, so t_i drops out identically.

5 Field solution at every order

5.1 Triangular scheme

At total momentum K, the order-r kernels (ν_r, A_r), with A_r the order-r part of H, solve $\mathcal{L}_K[\nu_r, A_r] + c^{(r)} = 0$. The operators are the same at every order:

$$L_N = 3\nu'' + K^2\nu - \tfrac{6}{t^2}\nu, \quad L_M = \tfrac{i}{2}K_j A'_{ji} - 2iK_i(\nu' - \nu/t), \quad L_T = \tfrac{1}{2}(A'' + \tfrac{2}{t}A') + \mathrm{TF}\, R_1[A] + 2\,\mathrm{TF}(K_iK_j)\nu.$$

Here $R_1[A]_{ij} = \frac{1}{2}(K^2A_{ij} - K_aK_iA_{aj} - K_aK_jA_{ai})$. The source $c^{(r)}$ is the multilinear coefficient with $(\nu_r, A_r) = 0$; it is explicit in the lower orders.

Write $c = K/\sqrt{3}$, $x = ct$, $u_1 = \sin x/x - \cos x$ and $u_2 = -\cos x/x - \sin x$. The coefficient c_r is the t^2 coefficient of (5) at order r. In particular $c_2 = [2(k_1^2 + k_2^2) - k_1 \cdot k_2]/9$ and $c_3 = \frac{2}{9}[-2\sum k_a^2 + \sum_{a<b} k_a \cdot k_b]$. Then

$$\nu_r = \frac{9c_r}{K^2}u_1(x) - \frac{1}{3c}\Big[u_2(x)\int_0^t u_1(cs)c_N^{(r)}\mathrm{d}s - u_1(x)\int_0^t u_2(cs)c_N^{(r)}\mathrm{d}s\Big], \tag{6}$$

$$h_S = \tfrac{3}{2K^2}\big[4K^2\nu_r + 12\nu_r'/t + 12\nu_r/t^2 + c_{\mathrm{Ham}}^{(r)}\big] \quad \text{(algebraic, from (Ham))}, \tag{7}$$

$$w = \tfrac{2i}{K^2}\int_0^t P_\perp c_M^{(r)}\mathrm{d}s, \qquad h = -2\int_0^t \tfrac{s}{t}\tfrac{\sin K(t-s)}{K}\,\mathrm{TT}[c_T^{(r)}](s)\,\mathrm{d}s, \tag{8}$$

$$A_r = (\hat{K}\hat{K} - \tfrac{1}{3}\delta)h_S + (K \otimes w + w \otimes K) + h. \tag{9}$$

At $K = 0$ no projector appears: $A_r = -2\int_0^t \frac{s}{t}(t-s)c_T^{(r)}(s)\mathrm{d}s$ and $\nu_r = c_r t^2 - \frac{1}{3}\int_0^t \frac{t^3 - s^3}{3ts}c_N^{(r)}\mathrm{d}s$. The particular parts are $O(t^4)$, so the free regular lapse mode is fixed *exactly* by (Ham) at $O(t^2)$.

Theorem 5 (Consistency at every order)**.** *The triangular solution satisfies all ten field equations at every order.*

Proof. Let $\mathbb{E}_{\mu\nu} = G_{\mu\nu} - 8\pi G T_{\mu\nu}$. Matter conservation is exact (Lemma 1), so $\nabla^\mu \mathbb{E}_{\mu\nu} = 0$. If all residuals vanish below order r, the order-r residual obeys the identity linearised about $a^2(-\mathrm{d}t^2 + \mathrm{d}x^2)$, with $\mathcal{H} = 1/t$. By construction:

- $\mathbb{E}^t{}_i = 0$, from (M);
- $\mathbb{E}^t{}_t = \frac{1}{2}\mathbb{E}$, from (N). With $\mathbb{E}^i{}_i = 3\tau$ this gives $\mathbb{E}^t{}_t = 3\tau$;
- for $K \neq 0$, the TT part of TF $\mathbb{E}^i{}_j$ vanishes.

Write the remaining trace-free part as $e_S(\hat{K}\hat{K} - \delta/3) + \hat{K} \otimes w + w \otimes \hat{K}$.

- The spatial component gives $K_j\mathbb{E}^j{}_i = 0$. Hence $w = 0$ and $\frac{2}{3}e_S + \tau = 0$.
- The time component gives $3\tau' + 6\tau/t = 0$. Regularity then forces $\tau = 0$, so $e_S = 0$ and (Ham) $\propto \mathbb{E}^t{}_t = 0$.

At $K = 0$, (T) holds in full, so $\tau = 0$. The spatial component gives $\partial_t\mathbb{E}^t{}_i + 4\mathcal{H}\mathbb{E}^t{}_i = 0$, so $\mathbb{E}^t{}_i = 0$ by regularity. □

5.2 Explicit kernels

First order [closed form].

$$\nu_1 = -(\sin x/x - \cos x), \qquad H_1 = -\tfrac{6}{k^2}\big(\tfrac{2\sin x}{x} - \cos x - 1\big)\big(k_ik_j - \tfrac{1}{3}\delta_{ij}k^2\big), \qquad \zeta_1 = 1.$$

Second order [closed form, general momenta]. Every integral above is of the form $\int_0^t B(s)\mathrm{d}s$, where $B = \sum_\omega e^{i\omega s} R_\omega(s)$ and each R_ω is a Laurent polynomial. The frequencies ω are integer combinations of $p/\sqrt{3}$, $q/\sqrt{3}$, $K/\sqrt{3}$ and K (with $p = |k_1|$, $q = |k_2|$, $K = |k_1 + k_2|$); they vanish on resonance surfaces. We integrate term by term in *entire* functions:

$$\int_0^t s^m e^{i\omega s}\mathrm{d}s = t^{m+1}\mathfrak{e}_m(i\omega t), \qquad \mathfrak{e}_m(z) = \int_0^1 u^m e^{zu}\mathrm{d}u \qquad (m \geq 0), \tag{10}$$

$$\int_0^t s^{-M}\Big[e^{i\omega s} - \sum_{j<M} \tfrac{(i\omega s)^j}{j!}\Big]\mathrm{d}s = J_M(\omega, t), \quad J_1 = -\operatorname{Cin}(\omega t) + i\operatorname{Si}(\omega t), \tag{11}$$

$$J_M = \tfrac{t^{1-M}E_M}{1-M} + \tfrac{i\omega}{M-1}J_{M-1}, \qquad E_M = e^{i\omega t} - \sum_{j<M} \tfrac{(i\omega t)^j}{j!}. \tag{12}$$

Here $\operatorname{Cin}(x) = \int_0^x (1 - \cos u)u^{-1}\mathrm{d}u$ is entire and even, and Si is entire and odd. The subtracted Taylor pieces sum to the principal part of B at $s = 0$, which vanishes (this is asserted in the code). *No division by any frequency occurs*, so the kernels are manifestly finite on every resonance surface:

- the scalar collinear surfaces $\mu = \pm 1$;
- the tensor and vector sound cone $\mu = -1/3$ (for $|k_1| = |k_2|$).

The removable $\ln|\alpha|$ terms that Ci would introduce never appear. The complete expressions for ν_2, h_S, w and h are in `solution/results/second_order_closed_forms.txt`. Each antiderivative is verified by exact differentiation. Substituting the assembled ν_2 back into the lapse equation (N) gives an exact zero in closed form. The scalar part h_S satisfies (Ham) by construction.

The momentum and trace-free equations follow from three facts: the verified antiderivatives, the verified Green functions, and Thm. 5. They are also confirmed independently by the exact series test through $O(t^8)$. A direct closed-form substitution into (M) and (T) is available (`--check all`) but was not run to completion: it takes hours, and it is redundant with these checks.

Third order [explicit nested integrals]. For $C^{(13)}$ only the configuration $(k, q, -q)$ is needed. The source $c^{(3)}$ is explicit (`solution/results/third_order_sources.txt`) in the first-order closed forms and in $K_2(k, q)$, $K_2(k, -q)$ and $K_2(q, -q)$. The last of these is the exact $K = 0$ kernel, never a limit. The third-order kernels are the formulas above applied to $c^{(3)}$, so they are double integrals with closed-form inner integrals.

6 The observable through third order

Proposition 6 (Tetrad)**.** *Fermi–Walker transport along u is parallel transport along ∂_t in $\tilde{g}$, that is, $\mathrm{d}E_a{}^i/\mathrm{d}t = -S^i{}_j E_a{}^j$. With $E(t_I) = \gamma(t_I)^{-1/2}$, the transported frame satisfies $E = \gamma_o^{-1/2} R$ at observation, where $R = 1 + \epsilon^2\Omega/2 + O(\epsilon^3)$ and Ω is antisymmetric. Consequently:*

- *$\Theta^{(1)}$ and $\Theta^{(2)}$ are those of the symmetric frame;*
- *$\Theta^{(3)}$ differs only by an infinitesimal rotation of $\Theta^{(1)}$;*
- *$C_\ell^{(22)}$ and $\operatorname{Re} C_\ell^{(13)}$ are individually unchanged.*

Proof. At first order $E_1' = -\frac{1}{2}\gamma_1'$, so $E_1 = -\frac{1}{2}\gamma_1$, which is the first-order part of $\gamma^{-1/2}$. The rotation acts on $a_{\ell m}^{(1)}$ by an anti-Hermitian generator, so $\operatorname{Re}\sum_m a_{\ell m}^{(1)*}(D(\Omega)a^{(1)})_{\ell m} = 0$ in every realisation. □

Take the background ray $\bar{X} = \hat{n}(t_o - t)$, $\bar{e} = -\hat{n}$, and write $X = \bar{X} + \epsilon X_1 + \epsilon^2 X_2/2$. All fields are evaluated on $(t, \bar{X})$, and $\sigma(a,b) = \sigma_{ij} a^i b^j$. Because σ, S and Γ vanish on the background, X_3 and e_3 never enter. The exact consequences of Lemma 2 are as follows.

Integrand.

$$\begin{aligned}
F_1 &= \sigma_{1,nn}, \qquad F_2 = \sigma_{2,nn} + 2X_1\cdot\partial\sigma_{1,nn} - 4\sigma_1(n,e_1),\\
F_3 &= \sigma_{3,nn} + 3X_1\cdot\partial\sigma_{2,nn} - 6\sigma_2(n,e_1) + 3X_2\cdot\partial\sigma_{1,nn} + 3(X_1\cdot\partial)^2\sigma_{1,nn}\\
&\quad - 12(X_1\cdot\partial)\sigma_1(n,e_1) - 6\sigma_1(n,e_2) + 6\sigma_1(e_1,e_1).
\end{aligned}$$

Ray. The displacements are

$$X_r = e_r(t_o)(t - t_o) + \int_t^{t_o} (t' - t) A_r(t')\mathrm{d}t',$$

with

$$\begin{aligned}
A_1 &= 2S_1 n - \Gamma_1(n,n) - n\sigma_{1,nn},\\
A_2 &= [2S_2 n - \Gamma_2(n,n) - n\sigma_{2,nn}] + 2(X_1\cdot\partial)[2S_1 n - \Gamma_1(n,n) - n\sigma_{1,nn}]\\
&\quad + 2[-2S_1 e_1 + 2\Gamma_1(n,e_1) + e_1\sigma_{1,nn} + 2n\sigma_1(n,e_1)].
\end{aligned}$$

The initial directions are the Taylor coefficients of $e(t_o) = -e^{-\zeta}\exp(-H/2)\,\hat{n}$, evaluated at $(t_o, 0)$.

Emission and endpoint. Let φ_r denote ν along the ray, with $\varphi_1 = \nu_1$, $\varphi_2 = \nu_2 + 2X_1\cdot\partial\nu_1$, and so on. Then

$$\begin{aligned}
&d_1 = -\bar{t}_\star\varphi_1, \qquad d_2 = \bar{t}_\star[u_1^2 - \varphi_2 - 2\varphi_1' d_1], \qquad u_1 = d_1/\bar{t}_\star,\\
&I_1 = \int F_1, \qquad I_2 = \int F_2 - 2F_1 d_1, \qquad I_3 = \int F_3 - 3F_1 d_2 - 3F_2 d_1 - 3F_1' d_1^2.
\end{aligned}$$

Clock. With $\omega(t) = \nu(t,0)$ and $W_r(T) = \int_0^T t(\omega_2 + \omega_1^2)_r\mathrm{d}t$, the observation-time shifts are

$$\begin{aligned}
D_1 &= [\bar{t}_I d_{I1} - W_1(\bar{t}_o) + W_1(\bar{t}_I)]/\bar{t}_o,\\
D_2 &= [\bar{t}_I d_{I2} + d_{I1}^2 - W_2(\bar{t}_o) - 2\bar{t}_o\omega_1(\bar{t}_o)D_1 + W_2(\bar{t}_I) + 2\bar{t}_I\omega_1(\bar{t}_I)d_{I1} - D_1^2]/\bar{t}_o.
\end{aligned}$$

The coordinate-event map Θ_c is then composed with the random observation time:

$$\Theta^{(2)} = \Theta_c^{(2)} + 2D_1\partial\Theta_c^{(1)}, \qquad \Theta^{(3)} = \Theta_c^{(3)} + 3D_2\partial\Theta_c^{(1)} + 3D_1\partial\Theta_c^{(2)} + 3D_1^2\partial^2\Theta_c^{(1)}.$$

Normalisation. Since $T \propto e^{-I}$, and $\langle\cdot\rangle_\Omega$ is the random sky mean:

$$\begin{aligned}
\Theta^{(1)} &= \langle I_1\rangle_\Omega - I_1, \qquad \Theta^{(2)} = I_1^2 - I_2 - 2\langle I_1\rangle_\Omega I_1 + 2\langle I_1\rangle_\Omega^2 - \langle I_1^2\rangle_\Omega + \langle I_2\rangle_\Omega,\\
\Theta^{(3)} &= -I_3 + 3I_1 I_2 - I_1^3 + 3\langle I_1\rangle_\Omega I_1^2 - 3\langle I_1\rangle_\Omega I_2 - 3\langle I_2\rangle_\Omega I_1 + 3\langle I_1^2\rangle_\Omega I_1 - 6\langle I_1\rangle_\Omega^2 I_1\\
&\quad + \langle I_3\rangle_\Omega - 3\langle I_1 I_2\rangle_\Omega + \langle I_1^3\rangle_\Omega + 6\langle I_1\rangle_\Omega^3 - 6\langle I_1\rangle_\Omega\langle I_1^2\rangle_\Omega + 6\langle I_1\rangle_\Omega\langle I_2\rangle_\Omega.
\end{aligned}$$

The dipole is removed last, by projection. It does not affect $C_{\ell\geq 2}$, and no deboost is implied.

Fourier dictionary. For a single mode a, with phase $e^{i\lambda k\cdot\hat{n}}$ and $\lambda = t_o - t$:

- $\sigma_{a,nn} = \frac{1}{2}Q'_a k^2(\mu_a^2 - \frac{1}{3})$;
- $\Gamma_a(n,n) = i[2(k\cdot n)n - k] + \frac{i}{2}[2(k\cdot n)H_a n - kH_{a,nn}]$;
- $e_a(t_o) = (1 + \frac{1}{2}H_a(t_o))\hat{n}$;
- $d_a = -\bar{t}_\star \nu_a(\bar{t}_\star)e^{i\lambda_\star k\cdot\hat{n}}$.

The second-order shear is $\sigma_{ab} = \frac{1}{2}H'_{ab} + \frac{1}{4}(H_aH_b + H_bH_a)' + H'_a + H'_b$, and

$$I_{ab} = \int_{\bar{t}_\star}^{t_o} \mathrm{d}t\, \{\sigma_{ab,nn} + i(k_b\cdot X_a)\sigma_{b,nn} + i(k_a\cdot X_b)\sigma_{a,nn} - 2\sigma_b(n,e_a) - 2\sigma_a(n,e_b)\} - [F_a d_b + F_b d_a]_{\bar{t}_\star},$$

$$\Theta_{ab} = I_aI_b - I_{ab} - \langle I_a\rangle_\Omega I_b - \langle I_b\rangle_\Omega I_a + 2\langle I_a\rangle_\Omega\langle I_b\rangle_\Omega - \langle I_aI_b\rangle_\Omega + \langle I_{ab}\rangle_\Omega + D_a\partial\Theta_b + D_b\partial\Theta_a.$$

The third-order kernel Θ_{abc} is the multilinear coefficient of the formulas above.

Independent generator. `observable_generator.py` rebuilds Θ *from the exact definitions* by fixed-point iteration in the multilinear algebra: the ray ODE, the emission root, the clock equation and the sky normalisation. It uses no hand-expanded formula, and it passes the Ward identities and the superhorizon check of §7 and §9.1.

7 Gauge invariance and Ward identities

Every coordinate and label is a physical scalar or frame: t, X, the surface $T = T_\star$, the fluid line $X = 0$, proper time from $T = T_I$, and $\gamma^{-1/2}$ in the labels. So $\Theta(\hat{n})$ is invariant under any diffeomorphism acting on fields and labels together. The only residual freedom is the conformal group of the primordial chart (§4). Its dilation generator must act as $\zeta \to \zeta + c \Leftrightarrow k \to e^{-c}k$. We verify this exactly:

- *Fields:* $K_2(k,0) = -k\cdot\partial_k K_1$ and $K_3(k,0,0) = (k\cdot\partial_k)^2 K_1$ solve every equation in closed form. Perturbed predictions fail.
- *Observable:* $\Theta_{sL} = -\kappa\partial_\kappa\Theta_s$ and $\Theta_{sLL} = (\kappa\partial_\kappa)^2\Theta_s$ hold through $O(\kappa^4)$, where κ scales the momenta. At this order the ray deflection, time delay, emission shift, clock, frame and normalisation all contribute.
- At first order, the result equals the independent Newtonian-gauge computation (SW, Doppler and ISW) for every ℓ.

8 The spectrum

8.1 Gaussian reduction

With $\mathcal{K}_r(k_1..k_r;\hat{n})$ the kernel of $\Theta^{(r)}$ (observer at the origin) and $A^{(r)}_{\ell m} = \int \mathrm{d}\Omega\, Y^*_{\ell m}\mathcal{K}_r$:

$$C_\ell^{(11)} = \tfrac{1}{2\ell+1}\textstyle\sum_m \int_k P|A^{(1)}_{\ell m}|^2, \qquad C_\ell^{(22)} = \tfrac{2}{2\ell+1}\sum_m \int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2,$$
$$C_\ell^{(13)} = \tfrac{3}{2\ell+1}\textstyle\sum_m \int_k\int_q P(k)P(q)A^{(1)}_{\ell m}(k)A^{(3)}_{\ell m}(k,q,-q)^*.$$

The disconnected part of 22 is proportional to $\delta_{\ell 0}$. In 13 all three pairings survive; the internal $(q,-q)$ pairing is a pairing of the fourth moment and is not removed by normal ordering. Odd orders vanish.

8.2 Monochromatic spectrum

After the complete observable is constructed, set

$$\mathcal{P}(k) = \sigma^2 \delta(\ln k/k_0), \qquad P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\delta(k-k_0), \qquad \langle g^2\rangle = \sigma^2 \text{ held fixed.}$$

The background and the labels $T_\star$, T_I and τ_o are also held fixed. Using joint rotation invariance, with $\hat{k} = \hat{z}$ and $\hat{q}_\mu = (\sqrt{1-\mu^2}, 0, \mu)$:

$$C_\ell^{(11)} = 4\pi\sigma^2 T_\ell(k_0)^2, \tag{13}$$

$$C_\ell^{(22)} = \frac{\sigma^4}{4\pi}\int_{-1}^{1}\mathrm{d}\mu \int \mathrm{d}\Omega \int \mathrm{d}\Omega'\, P_\ell(\hat{n}\cdot\hat{n}')\, \mathcal{K}_2(\hat{n})\mathcal{K}_2(\hat{n}')^*, \tag{14}$$

$$C_\ell^{(13)} = \frac{3\sigma^4}{2}\, i^\ell T_\ell(k_0) \int_{-1}^{1}\mathrm{d}\mu \Big[\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, \mathcal{K}_3(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu; \hat{n})\Big]^*. \tag{15}$$

All sky integrals in closed form. Every term of $\mathcal{K}_r$ has the form $c(\lambda,\ldots)\,\Pi(\hat{n})\, e^{i\hat{n}\cdot V}$. Here $V = \sum_a k_a\lambda_a$ collects the line-of-sight and nested times, and Π is a polynomial. By Funk–Hecke and $n_i \to -i\partial_{V_i}$:

$$\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\,\Pi(\hat{n}) e^{i\hat{n}\cdot V} = \Pi(-i\partial_V)\big[4\pi i^\ell j_\ell(|V|) P_\ell(\hat{z}\cdot\hat{V})\big],$$

$$\iint P_\ell(\hat{n}\cdot\hat{n}')\Pi(\hat{n}) e^{i\hat{n}\cdot V}\Pi'(\hat{n}') e^{-i\hat{n}'\cdot V'} = \Pi(-i\partial_V)\Pi'(i\partial_{V'})\big[(4\pi)^2 j_\ell(|V|) j_\ell(|V'|) P_\ell(\hat{V}\cdot\hat{V}')\big].$$

The sky means in the normalisation terms are the $\ell = 0$ case, $4\pi j_0$. The remaining integrals are therefore:

- for $C^{(22)}$: μ, plus per copy one line-of-sight time, one ray time and one field-kernel time;
- for $C^{(13)}$: μ and up to five nested times.

All integrands are elementary or closed-form.

Theorem 7 (Existence)**.** *For all finite $0 < x_\star < x_o$, the integrals for $C_\ell^{(22)}$ and $C_\ell^{(13)}$ converge absolutely. The restriction to the shell is well defined.*

Proof. The domains are compact. The integrands are bounded for the following reasons:

- the field kernels are continuous in time and momenta, and finite on resonances, by the entire-function representation;
- $c_M^{(2)}$ is analytic and vanishes at $K = 0$, so $c_M = O(K)$, and the projector terms $(\hat{K}\hat{K} - \delta/3)h_S$ and $K \otimes w$ are bounded near $K \to 0$. They are discontinuous only at $\mu = -1$, a null set;
- $K_2(q, -q)$ in $C^{(13)}$ is the exact $K = 0$ kernel;
- the ray integrals are over finite intervals.

□

9 Limits

9.1 Superhorizon limit, exact in ζ

Theorem 8. *Let all momenta scale as $\kappa \to 0$ at fixed times. Then, to all orders in ϵ,*

$$\Theta(\hat{n}) = -\tau^2\, e^{-2\zeta}\ \mathrm{TF}\left[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\right]_o \hat{n}^i\hat{n}^j + O(\kappa^3), \qquad \tau^2 = \tfrac{1}{6}(\bar{t}_o^2 - \bar{t}_\star^2).$$

Proof. $\sigma = O(\kappa^2)$, while its spatial variation along the ray, the direction corrections and the ray displacement all bring at least one more power of κ. The endpoint and clock shifts are themselves $O(\kappa^2)$ and multiply $O(\kappa^2)$ quantities. Hence $I = \frac{1}{2}\left[(e^H)_{nn}\right]_{\bar{t}_\star}^{t_o}$ at $X = 0$. Here $e^{-2\zeta}\sigma = \frac{1}{2}(e^H)'$ and $e^\zeta e$ is $\exp(-H/2)\hat{n}$ to this order. Inserting (5) gives I. Since I is a pure quadrupole, $\langle I\rangle_\Omega = 0$, and $\Theta = -I + O(\kappa^4)$.

The bound is uniform in the angles, because the remainders are bounded (projectors are bounded), so the expansion may be taken inside the compact integrals.

The independent generator confirms the formula at first and second order, at (k, q) with general momenta. The corresponding third-order run at $(k, q, -q)$ was *not completed* (§10). At third order the formula rests on two things: the proof above, which uses only the gradient-expansion data (5) (exact in ζ and verified for arbitrary ζ), and the generator's third-order Ward identity W3. □

With $\Xi \equiv x_o^2 - x_\star^2 = k_0^2(\bar{t}_o^2 - \bar{t}_\star^2)/3$, exact Wick moments of $(g, \partial g, \partial\partial g)$ at a point give the leading $O(k_0^4)$ results [closed form]:

$$C_2^{(11)} = \frac{4\pi}{225}\sigma^2\Xi^2, \qquad C_2^{(22)} = \frac{272\pi}{675}\sigma^4\Xi^2, \qquad C_2^{(13)} = \frac{16\pi}{75}\sigma^4\Xi^2,$$

$$C_2 = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2\left[1 + \frac{29}{3}\epsilon^2\sigma^2\right] + O(\epsilon^6) + O(k_0^6), \qquad C_{\ell\geq 3} = O(k_0^6).$$

Two checks: $C_2^{(11)}$ agrees with the line-of-sight transfer function $T_2 \to -\frac{2}{15}k_0^2\tau^2$, and the direct ϵ-expansion of $\langle\sum_m |a_{2m}|^2\rangle$ reproduces $\frac{1}{4}C^{(22)} + \frac{1}{3}C^{(13)}$. The fourth-order correction is a finite, positive renormalisation of the quadrupole by the factor $1 + \frac{29}{3}\epsilon^2\sigma^2$. It is independent of T_I at this order.

9.2 Late time, first order

Proposition 9. *For $\ell \geq 2$ and fixed $x_\star$, $T_\ell(x_\star, x_o) = O(\ln x_o/x_o)$ as $x_o \to \infty$. Hence $C_\ell^{(11)} \to 0$.*

Proof. Write $T_\ell = -3\int_0^{x_o - x_\star} s'(x_o - u)w_\ell(\sqrt{3}u)\mathrm{d}u$, with $w_\ell = j_\ell/3 + j_\ell''$. Two integrations by parts give $\int_0^\infty e^{iu}w_\ell(\sqrt{3}u)\mathrm{d}u = 0$ exactly for $\ell \geq 2$, because $j_\ell(0) = j_\ell'(0) = 0$. This is the observer-local sound-cone cancellation. Since $s'(x) = \sin x + O(1/x)$, the $O(1)$ oscillation therefore vanishes. The remaining pieces are oscillatory with distinct frequencies $\sqrt{3} \pm 1$ and amplitudes $O(1/x_o)$, giving $O(\ln x_o/x_o)$ by integration by parts. □

At fourth order the late-time limit is non-uniform at the resonances $\mu = \pm 1$ and $\mu = -1/3$. Only the order "angle first" is admissible. Its evaluation is [open].

9.3 General spectra

Theorem 10 (Infrared)**.** *For spectra smooth on $(0,\infty)$, consider the $q \to 0$ region. By the Ward identities, $K_2(k,q) = -k\cdot\partial_k K_1(k) + O(q)$ and $K_3(k,q,-q) = (k\cdot\partial_k)^2 K_1(k) + O(q^2)$. Here the kernels are analytic in q at fixed finite times, and the absence of an $O(q)$ term in K_3 follows from its symmetry under $q \to -q$. The $O(q)$ cross term in $|K_2|^2$ is odd in $\hat{q}$ and integrates to zero.*

Both the q-long and the k-long regions of $C^{(22)}$ contribute. Let $\sigma_L^2 = \int_{q<\Lambda} P$ and $D = k\cdot\partial_k$. The infrared-sensitive part of $\frac{1}{4}C^{(22)} + \frac{1}{3}\operatorname{Re} C^{(13)}$ is then

$$\sigma_L^2 \,\frac{1}{2\ell+1}\int_k P(k)\Big[\sum_m |DA^{(1)}_{\ell m}|^2 + \operatorname{Re}\sum_m A^{(1)}_{\ell m} D^2 A^{(1)*}_{\ell m}\Big] = \frac{\sigma_L^2}{2}\,\frac{1}{2\ell+1}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\Big[\sum_m |A^{(1)}_{\ell m}|^2\Big](D+3)^2 P(k).$$

This vanishes identically for scale-invariant $P \propto k^{-3}$. Otherwise it is the physical rescaling of the short-mode spectrum by the long-mode variance. The remainder is $O(q^2)P(q)$, which is integrable for $\mathcal{P}(q)$ with $\int_0 \mathcal{P}(q) q^2 \,\mathrm{d}\ln q < \infty$.

Ultraviolet [power counting only]. At fixed finite times:

- the first-order kernels are bounded uniformly in k;
- each order of the sources carries at most two spatial derivatives;
- each ray displacement carries one.

Counting powers on this basis gives $|\mathcal{K}_2(k,q)| \lesssim q^2$ and $|\mathcal{K}_3(k,q,-q)| \lesssim q^4$ as $q \to \infty$. That suggests absolute convergence of $C^{(22)}$ and $C^{(13)}$ whenever $\int^\infty \mathcal{P}(q) q^4 \,\mathrm{d}\ln q < \infty$.

This is *not* a proof. A rigorous bound needs uniform estimates of the lapse Green function near $s \to 0$ (where $|G_N|$ grows like $1/(K^3 t s)$) against sources whose small-s Taylor coefficients grow with q. The true large-q behaviour for a scale-invariant spectrum, including possible cancellations, is [open].

10 Verification ledger

All checks are exact symbolic computations (SymPy 1.14). None uses floating point or quadrature as evidence. "Generic 2-jets" means that every function value and derivative is an independent symbol, so a zero residual is a proof.

Check	Script and scope
Lemmas L1–L5; generic field equations	`verification/check_exact_lemmas.py`, `check_fw_conformal.py`, `check_field_equations_generic.py` (generic 2-jets)
Mixed trace-free form (T)	`solution/check_evolution_form.py` (generic 2-jets)
First order, two gauges, all ℓ	`verification/check_first_order.py`; `solution/first_order_engine_check.py`
Closed form T_2, T_3	`verification/transfer_closed_form.py`
Green functions, $K \to 0$	`solution/check_green.py`

Check	Script and scope
2nd-order sources and operators	`solution/second_order.py`
2nd order, all 10 equations	`solution/second_order_consistency.py`: exact Taylor series through $O(t^8)$, general K and $K = 0$
2nd-order closed forms	`solution/closed_second_order.py`: every antiderivative by exact differentiation; (N) by closed-form substitution (`--check N`); (M) and (T) via the theorem and the series test
3rd-order sources and operators	`solution/third_order.py`
3rd order, all 10 equations	`solution/fast_series.py 8`: the pair kernels, including the exact $K = 0$ pair, then the triple at $(k, q, -q)$ with general momenta. The *full nonlinear* equations are recomputed with the solved kernels and vanish through $O(t^6)$. The arithmetic is exact (sparse polynomials over $\mathbb{Q}(i)$ with formal inverse momenta; denominators are cleared exactly at the zero test). Negative controls (a wrong c_3, a corrupted $K = 0$ pair kernel) fail at the expected orders. The run takes about 70 s. `third_order_consistency.py` is the slower SymPy reference implementation of the same test.
Dilation Ward identities (fields)	`solution/check_ward_dilation.py`, `check_ward_dilation3.py` (closed form; negative controls)
Observable expansion formulas	`solution/check_observable_expansion.py` (arbitrary fields, general jets)
Observable Ward identities	`solution/observable_generator.py W2|W3` through $O(\kappa^4)$
Superhorizon observable	`solution/observable_generator.py SH`: orders 1 and 2, general (k, q). *Not completed:* `SH3` (third order at $(k, q, -q)$). Its fields are built by the fast engine in about 12 s, but the generator's SymPy assembly of the observable did not finish within 50 minutes. Porting the generator to the fast arithmetic would remove this; it was not done.
Superhorizon C_2	`solution/superhorizon.py`
Normalisation; Wick and Fourier measures	`verification/normalization_expansion.py`; `validation/check_algebra.py`

11 What remains unevaluated

1. For general k_0, $C_\ell^{(22)}$ and $C_\ell^{(13)}$ are given as exact absolutely convergent integrals: over μ and the nested times, with closed-form integrands, and with all sky integrals evaluated. We do not expect, and did not find, a closed form. They are *not* evaluated beyond the superhorizon limit.

2. The late-time limit at fourth order (non-uniform at the resonances) is not evaluated.

3. Ultraviolet convergence for spectra without compact support is established by power counting only, not by a rigorous bound.

4. The infrared statement (Thm. 10) assumes that the kernels are analytic in q at $q = 0$ at fixed finite times. This holds for the integral representations by construction, but the uniformity of the expansion in k over unbounded spectra is not proved.

5. The perturbative domain is treated order by order: a single emission intersection and no caustics. There is no statement about convergence in ϵ.